A Capstone Project Presented to
College of Information and Communications Technology
Catanduanes State University

Gamifying Cultural Exploration:

Enhancing User Engagement Through Augmented Reality (AR) in Bato, Catanduanes Heritage Sites

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Development Team

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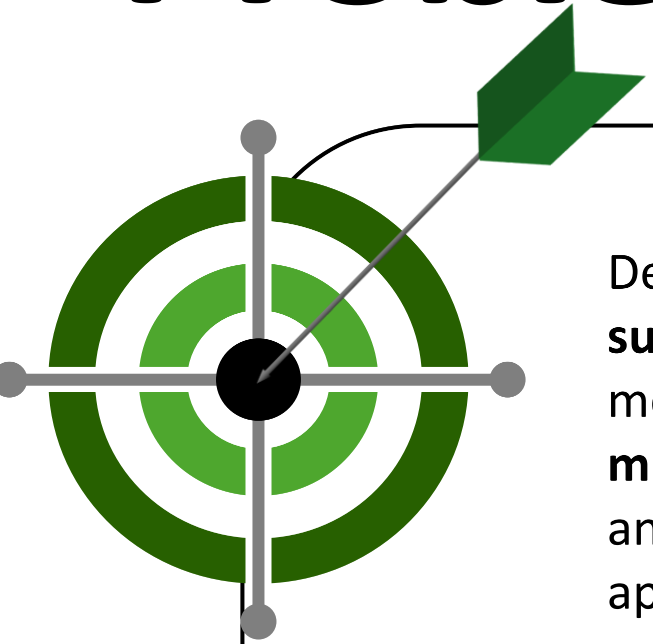
The fourth-year BSIT students behind the development of ARIBA BATO

Problem Statement



Despite the rising number of visitors, tourism experiences in Bato remain largely **surface-level** due to the absence of immersive and interactive interpretive mechanisms. Existing tourism practices prioritize visitor attraction but **offer minimal support for cultural education** and meaningful engagement, resulting in an “experience deficit” where increased visitation does not translate into cultural appreciation, sustained tourist interest, or long-term tourism value.

This imbalance between visitor volume and cultural engagement underscores the need to address how tourism experiences in Bato, Catanduanes can be enhanced to promote greater awareness, appreciation, and sustainable development of its cultural heritage.

Proposed Solution / Objectives

Identify and document significant cultural sites, practices, and historical narratives in Bato, Catanduanes, that can be integrated into a gamified AR engine.

Design and develop an AR mobile application incorporating gamification features, including quests, rewards, and interactive storytelling, for cultural exploration.

Assess user engagement and the perceived impact of the gamified AR application by evaluating user motivation, interaction, enjoyment, and levels of cultural awareness and learning experienced during mobile cultural exploration.

Evaluate the usability of the AR application using the System Usability Scale (SUS), measuring ease of use, learnability, functionality, overall user satisfaction and system complexity.

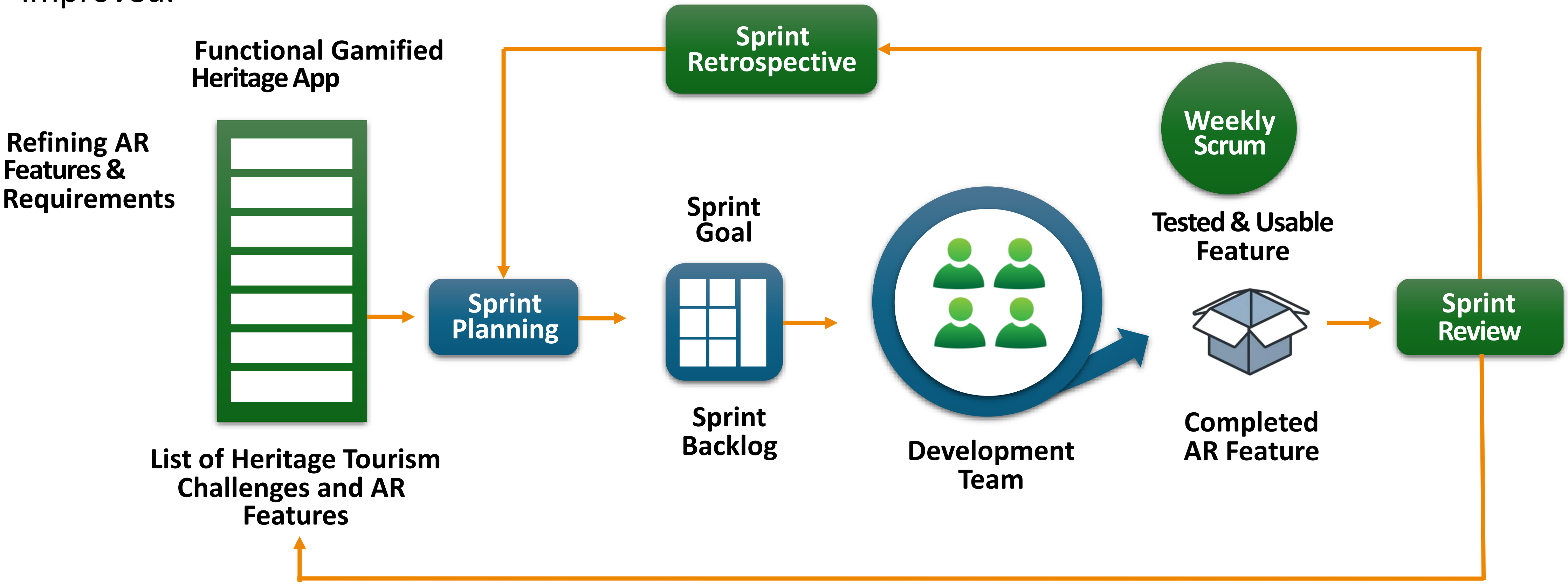
Statement of Novelty

This is the **first mobile application** specifically created for the heritage sites of Bato, Catanduanes, using Augmented Reality (AR) and game features to change how people visit these historical locations. By moving away from traditional methods, the project allows visitors to see history come to life through 3D digital items shown on their phone screens at places like the Bato Church and the Shrine of Batalay.

With fun activities such as treasure hunts, narrated stories, and quizzes with rewards, the app turns a simple visit into an interactive learning game. This new approach helps younger, tech-savvy visitors connect more deeply with local history and helps preserve the culture of the area in a modern way.

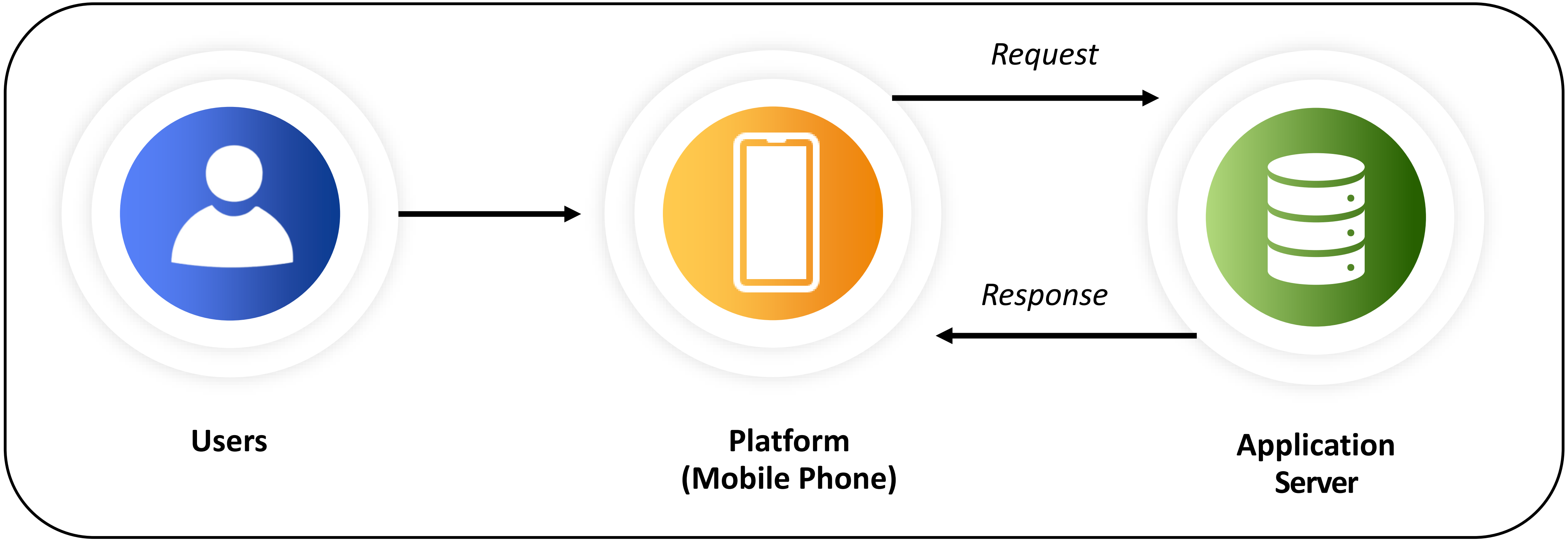
Methodology

The **Agile Scrum** framework was adapted to guide the development of the Gamified Augmented Reality (AR) Heritage Tourism Application. The Scrum model provides an iterative and incremental approach to software development, ensuring that progress is continuously reviewed and improved.

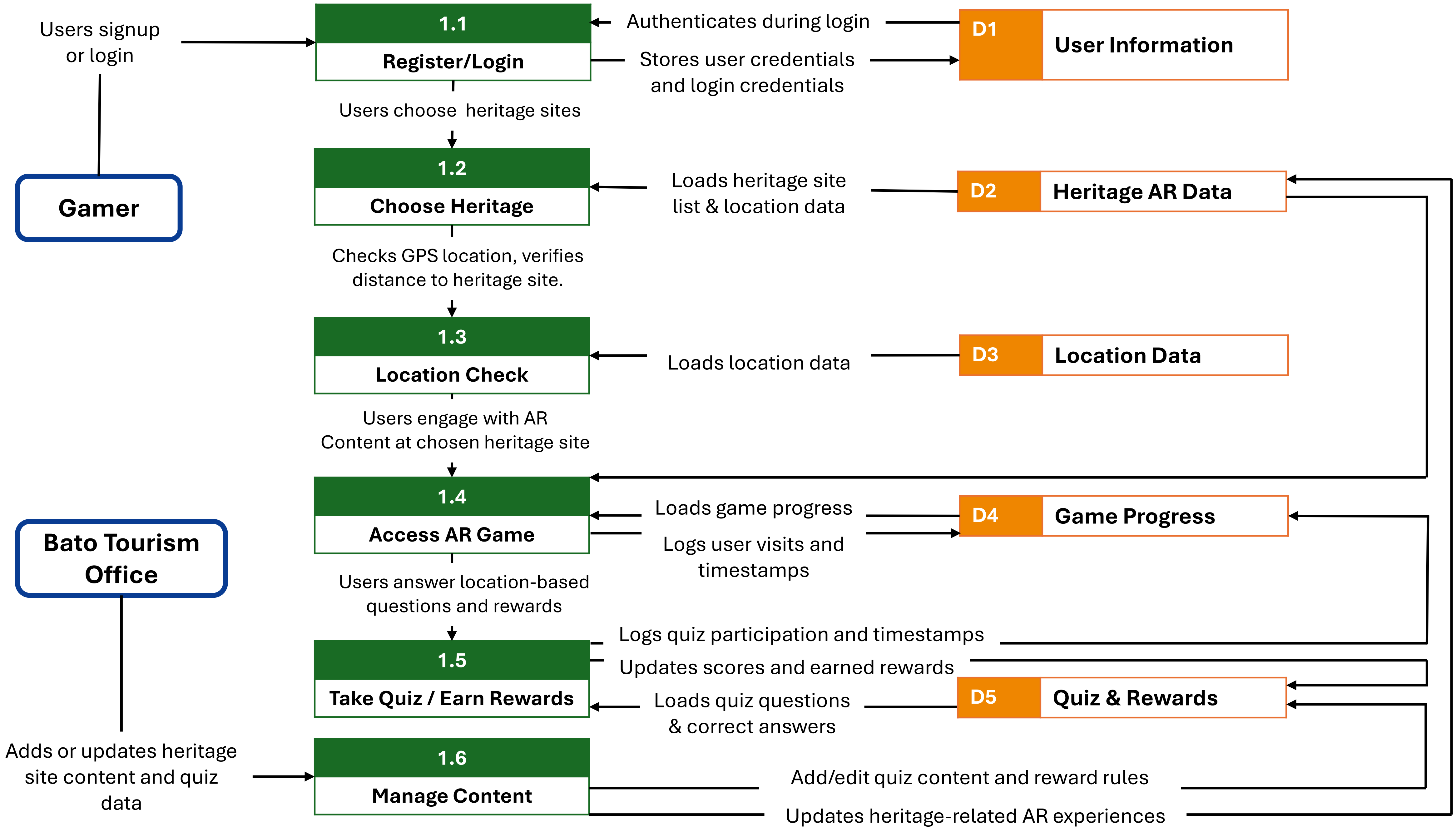


System Architecture

The system follows a **client–server architecture** in which users access the AR application through their mobile devices. The application server processes user requests, handles system logic, and manages database operations, then sends responses back to the mobile device.



Data Flow Diagram

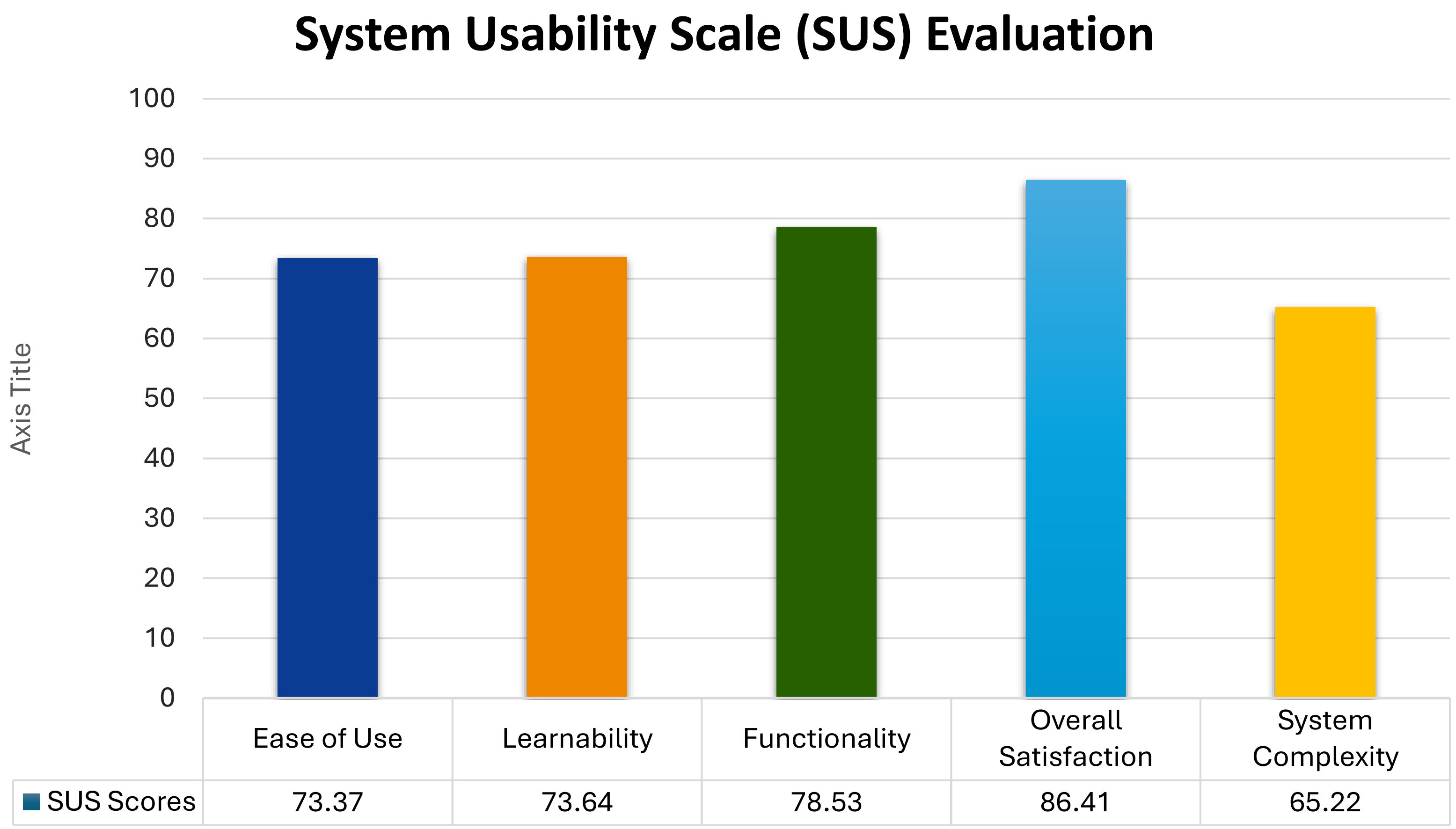

Users signup or login
Gamer
Bato Tourism Office
Adds or updates heritage site content and quiz data
1.1
Register/Login
Authenticates during login
Stores user credentials and login credentials
D1
User Information
Users choose heritage sites
1.2
Choose Heritage
Loads heritage site list & location data
D2
Heritage AR Data
Checks GPS location, verifies distance to heritage site.
1.3
Location Check
Loads location data
D3
Location Data
Users engage with AR Content at chosen heritage site
1.4
Access AR Game
Loads game progress
Logs user visits and timestamps
D4
Game Progress
Users answer location-based questions and rewards
1.5
Take Quiz / Earn Rewards
Logs quiz participation and timestamps
Updates scores and earned rewards
Loads quiz questions & correct answers
D5
Quiz & Rewards
1.6
Manage Content
Add/edit quiz content and reward rules
Updates heritage-related AR experiences

System Evaluation Method

The research used the **System Usability Scale (SUS)** by Brooke (1986) to measure usability of the augmented reality application. The SUS consists of ten standardized items rated on a five-point Likert scale, and they were chosen to address ease of use, learnability, functionality, and satisfaction.

1. I thought the system was easy to use.
2. I found the system very difficult to use.
3. I would imagine that most people would learn to use this system very quickly
4. I needed to learn a lot of things before I could get going with this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I think that I would like to use this system frequently.
8. I felt very confident using the system.
9. I found the system unnecessarily complex.
10. I think that I would need the support of a technical person to be able to use this system.

Results & Discussion

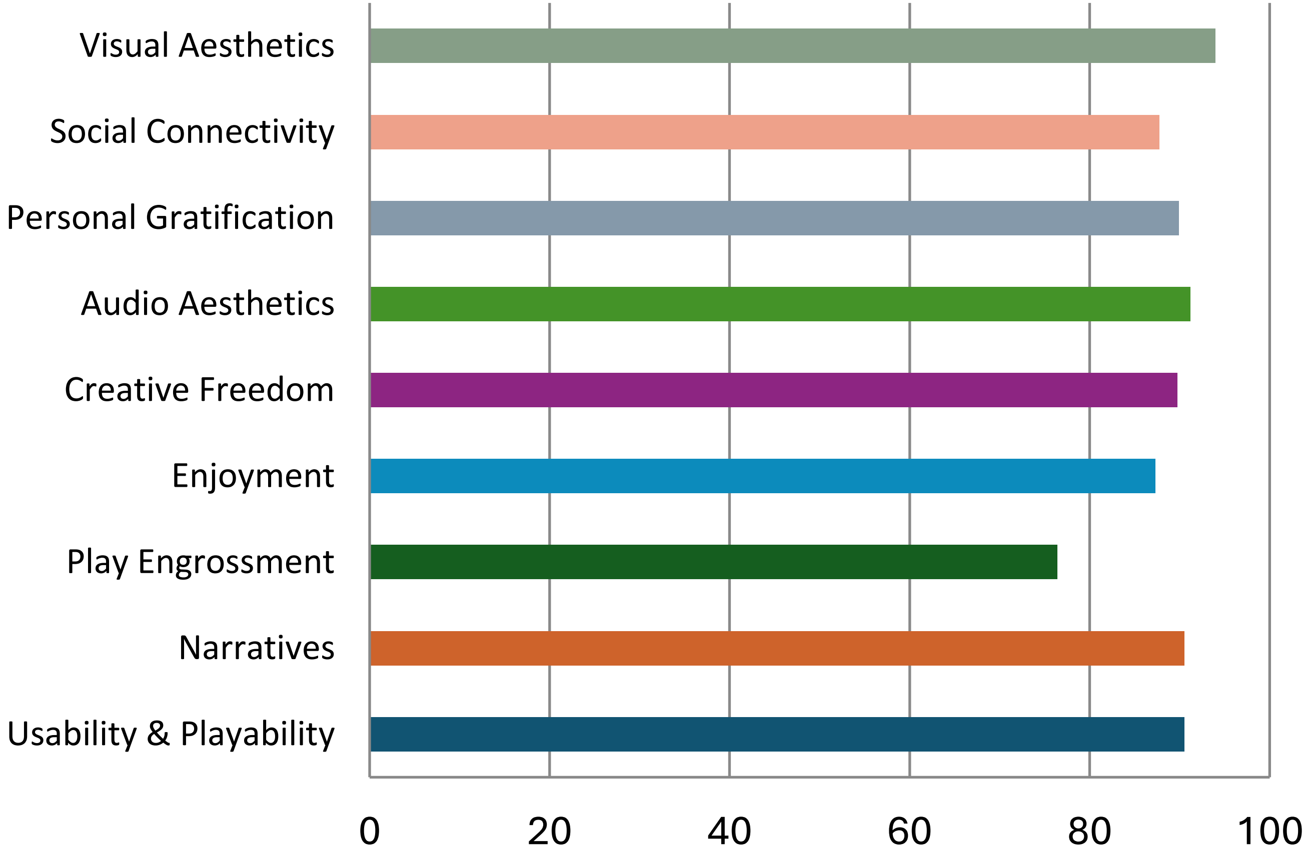


Category	Grade / Rating
Overall SUS	B- / Good
Ease of Use	B- / Good
Learnability	B- / Good
Functionality	B+ / Good
Overall Satisfaction	A+ / Best Imaginable
System Complexity	C / Ok

Results & Discussion

Game User Experience Satisfaction Scale

GUESS-18 Category Scores



Overall GUESS Score: 87.28%

High level of game user experience satisfaction indicating strong engagement and enjoyment.

Category	Score (%)
Usability & Playability	90.53
Narratives	90.53
Play Engrossment	76.40
Enjoyment	87.27
Creative Freedom	89.75
Audio Aesthetics	91.15
Personal Gratification	89.91
Social Connectivity	87.73
Visual Aesthetics	93.94

Recommendations

01

Future researchers are encouraged to **expand** the identification and documentation of cultural sites, practices, and historical narratives in Bato, Catanduanes, including additional heritage locations, oral histories, and lesser-known traditions to enrich the gamified AR content.

02

Future studies may focus on **improving the AR mobile application** by introducing advanced gamification features, optimizing AR performance, refining interface design, and ensuring cross-device compatibility to enhance the cultural exploration experience.

Recommendations

03

Anchored on the 6Ps of Research Outputs:

a. **Publication** – Results may be published in journals or presented at conferences on AR, gamification, cultural heritage, and tourism technology.

b. **Patent** – The gamified AR framework may be considered for intellectual property registration as a novel heritage exploration approach.

c. **Product** – The application may be developed into a full-scale digital tourism tool for Bato and other municipalities.

d. **People Services** – Training programs may be conducted for tourism officers, educators, and local guides to ensure effective use.

e. **Places and Partnerships** – Collaborate with local governments, schools, cultural organizations, and tourism stakeholders to enrich content and promote adoption.

f. **Policies** – Institutional and local policies may support sustainable integration of AR tools in cultural tourism.

System Walkthrough

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Thank You !